

Moller scattering 3

The Casimir trick uses matrix arithmetic to sum over spin states.

$$\begin{aligned}\sum_{abcd} \mathcal{M}_t \mathcal{M}_t^* &= \frac{e^4}{t^2} \text{Tr} \left((\not{p}_4 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu \right) \text{Tr} \left((\not{p}_3 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right) \\ \sum_{abcd} \mathcal{M}_t \mathcal{M}_u^* &= \frac{e^4}{tu} \text{Tr} \left((\not{p}_4 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu (\not{p}_3 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right) \\ \sum_{abcd} \mathcal{M}_u \mathcal{M}_u^* &= \frac{e^4}{u^2} \text{Tr} \left((\not{p}_3 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu \right) \text{Tr} \left((\not{p}_4 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right)\end{aligned}$$

In Mandelstam variables

$$\begin{aligned}\text{Tr} \left((\not{p}_4 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu \right) \text{Tr} \left((\not{p}_3 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right) &= 8s^2 + 8u^2 + 64m^2t - 64m^4 \\ \text{Tr} \left((\not{p}_4 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu (\not{p}_3 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right) &= -8s^2 + 64m^2s - 96m^4 \\ \text{Tr} \left((\not{p}_3 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu \right) \text{Tr} \left((\not{p}_4 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right) &= 8s^2 + 8t^2 + 64m^2u - 64m^4\end{aligned}$$

Recall that slash matrices are formed as

$$\not{p} = p^\mu g_{\mu\nu} \gamma^\nu = p^0 \gamma^0 - p^1 \gamma^1 - p^2 \gamma^2 - p^3 \gamma^3$$

Eigenmath script